04: Prediction for SLR

Stat 230 - Spring 2026

1 Warm-up questions

Is education level associated with income? Researchers collected education level (in years) and income (in thousands of dollars) for a random sample of 32 employees working for the city of Riverview.

The researchers fit a simple linear regression of income on education and obtained the following output:

term	estimate	std.error	statistic	p.value
(Intercept)	11.321	6.123	1.849	0.074
education	2.651	0.370	7.173	<0.001

Q1. Write the fitted regression equation.

Q2. Interpret the slope coefficient in the context of the problem. (Don't forget to specify units.)

Q3. Interpret the intercept in the context of the problem. (Don't forget to specify units.)

A new researcher joined the team and decided that education should be standardized (to have mean 0 and SD 1) before fitting the regression model. The output from this regression is shown below:

term	estimate	std.error	statistic	p.value
(Intercept)	53.742	1.587	33.861	<0.001
scale(education)	11.567	1.613	7.173	<0.001

Q4. Write the fitted regression equation for this new model.

Q5. Interpret the slope coefficient in the context of the problem. (Don't forget to specify units.)

Q6. Interpret the intercept in the context of the problem. (Don't forget to specify units.)

Q7. Compare the two models. What's the same? What's different?

2 Prediction and confidence intervals

> Tip
>
> The R code for the following questions is found at
> https://aluby.github.io/stat230/activities/04R-prediction
> The URL is also posted on Moodle.

Now, we'll explore the `city` data using R.

Q1. Make a plot of `income` against `education` and overlay the linear regression equation. Does it look like what you expected?

Q2. Make a plot of `income` against `scale(education)` and overlay the linear regression equation. Does it look like what you expected?

Q3. Find the linear regression equation from Q1. Confirm that it matches the table at the beginning of this worksheet.

Q4. The first person in this dataset has `education = 8`. Calculate the expected income of this person using the first regression equation.

Q5. If we want to predict the income of **this** person, should we use a confidence interval or a prediction interval?

Q6. Use R to construct the appropriate 89% interval for this person's income. Record this interval below.

Q7. Interpret the interval in context.

Q8. Run the code to produce a scatterplot, regression line, and both types of intervals. Which is the prediction interval and which is the confidence interval? How can you tell?